

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Results-driven **Software Quality Engineer / Developer Tools Engineer** with **6+ years of experience** designing scalable test infrastructure, automation frameworks, systems-integration validation, developer tooling, and release-readiness solutions for enterprise, device, and distributed software platforms.
- Strong programming expertise in **Python, C, C++, Rust, Java, and TypeScript**, with hands-on experience developing production-grade automation utilities, API validation tools, firmware test harnesses, orchestration services, and debugging workflows.
- Experienced designing unified test frameworks covering **unit, integration, system, regression, and end-to-end testing**, with reproducible execution, isolated test environments, automated failure triage, flake detection, and signal-quality guardrails.
- Hands-on experience integrating automated test infrastructure with **GitHub, Buildkite, CI/CD pipelines, artifact management, release gates, staging/production promotion criteria, automated regression filing, and traceable release artifacts**.
- Strong systems engineering and Linux expertise including **processes, networking, storage, udev, systemd, device provisioning, telemetry capture, log analysis, containerization, Docker, and hardware-software integration**.
- Experienced designing and supporting **hardware-in-the-loop (HIL) environments**, device orchestration, automated flashing, provisioning, API/firmware validation, telemetry collection, rig health monitoring, and repeatable hardware validation workflows.
- Proven ability to collaborate with embedded software, systems engineering, release infrastructure, QA, vendor technicians, and Product teams while communicating quality metrics, pass rates, release risks, debugging findings, and rollback/forward recommendations.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Developer Tooling:** Python, C, C++, Rust, Java, TypeScript, Bash/Shell, REST APIs, CLI Tools, Automation Utilities, Firmware Validation Tools
- **Test Infrastructure:** Unit Testing, Integration Testing, System Testing, End-to-End Testing, Regression Infrastructure, Unified Test Frameworks, Reproducible Test Runs, Isolated Test Environments
- **Hardware-in-the-Loop:** HIL Labs, Device Orchestration, Automated Flashing, Provisioning, Telemetry Capture, Rack Validation, Power/Networking Systems, Device Test Automation
- **CI/CD & Releases:** Buildkite, GitHub, Jenkins, GitHub Actions, CI/CD Pipelines, Release Gates, Artifact Management, Staging/Production Promotion, Release Notes, Rollback/Forward Strategies
- **Linux & Systems:** Linux, Processes, Networking, Storage, udev, systemd, Device Management, Log Capture, Kernel/System Diagnostics, Docker, Containerized Test Environments
- **Quality Engineering:** Flake Detection, Flaky Test Quarantine, Failure Triage, Root Cause Analysis, Pass-Rate Monitoring, Stability Metrics, Release Readiness, Quality Signals, Replayable Bug Reports
- **Collaboration & Tools:** GitHub, Linear, Slack, Jira, Vendor Enablement, QA Lab Procedures, Embedded Software Testing, Firmware Debugging, Systems Integration, Release Infrastructure

EXPERIENCE

Test Automation Engineer *PCMI LLC (Full Time)*

Oct 2024 – Present
San Francisco, CA, USA

- Designed and maintained a scalable **unified test infrastructure** supporting unit, integration, system, regression, and end-to-end validation with reproducible execution, reusable fixtures, isolated environments, and consistent quality signals across software releases.
- Developed Python, C++, TypeScript, and shell-based developer tools for API validation, system health checks, log capture, automated diagnostics, result triage, test-data preparation, and replayable bug-report generation.
- Integrated automated suites with **GitHub and Buildkite CI/CD pipelines**, implementing release gates, staging and production promotion criteria, artifact collection, automated regression reporting, and traceable release-validation results.
- Built failure-triage and test-reliability workflows that analyzed pass rates, recurring failures, timing behavior, and historical execution data to detect flaky tests, quarantine unstable scenarios, and route failures to appropriate engineering owners.
- Created dashboards and automated alerts for **test pass rates, stability, regression health, flaky-test trends, and release readiness**, providing engineering teams with actionable signals before deployment decisions.
- Worked with Linux-based systems and containerized environments to troubleshoot processes, networking, storage, systemd services, device connectivity, resource usage, and Docker-based test dependencies during integration and release validation.
- Partnered with developers and release-infrastructure engineers to improve testability, debug complex cross-service failures, define staged rollout criteria, and validate rollback/forward strategies for high-risk production releases.
- Produced developer-facing documentation, runbooks, debugging procedures, test-execution guides, failure classifications, and repeatable operational workflows while communicating release risks and root-cause findings to engineering and Product stakeholders.

Environment: Python, C++, TypeScript, Bash, Linux, GitHub, Buildkite, Jenkins, Docker, REST APIs, CI/CD, Unit Testing, Integration Testing, System Testing, End-to-End Testing, Regression Testing, Artifact Management, Release Gates, Flake Detection, Failure Triage, Log Capture, Release Readiness, Jira, Agile

Environment: Playwright, JavaScript, TypeScript, Selenium WebDriver, Postman, REST Assured, JSON, YAML, SQL, PostgreSQL, MySQL, Oracle, GitHub, GitHub Actions, GitHub Workflows, Docker, Snyk, Wiz, Jira, Agile Scrum, Test Planning, Unit Testing, Feature Testing, Regression Testing, Smoke Testing, Risk-Based Testing, Code Coverage, API Testing, End-to-End Testing

SDET

Dec 2023 – Sept 2024
Remote, USA

Client: CVS

- Designed and implemented test infrastructure for distributed healthcare applications, supporting unit, integration, system, regression, API, and end-to-end validation through reliable and repeatable automated workflows.
- Developed Python, C++, TypeScript, and API-based automation utilities for service validation, environment setup, telemetry collection, log capture, debugging, test-data generation, and automated result analysis.
- Integrated automated validation with **GitHub, Buildkite/Jenkins, and CI/CD release pipelines**, enforcing quality gates, capturing test artifacts, publishing execution results, and supporting staging and production release-readiness decisions.
- Built automated failure-triage workflows to correlate test failures with logs, application telemetry, service responses, and infrastructure events, enabling faster root-cause analysis and creation of replayable defect reports.
- Implemented quality dashboards and alerts for regression pass rate, service stability, flaky-test frequency, release health, and recurring failure patterns; quarantined unstable tests until root causes were addressed.
- Performed deep debugging of distributed and Linux-based applications across software services, APIs, networking, storage, processes, Docker containers, and cloud infrastructure to isolate integration and release failures.
- Collaborated with Software Engineers, Platform teams, QA, DevOps, Product, and release stakeholders to improve testability, define promotion criteria, validate staged deployments, and establish rollback or forward-fix strategies.
- Created test procedures, runbooks, release documentation, triage guides, and standardized execution workflows that enabled engineers and external QA teams to perform consistent validation and report trustworthy results.

Environment: Python, C++, TypeScript, Linux, GitHub, Buildkite, Jenkins, Docker, REST APIs, CI/CD, Distributed Systems, Telemetry, Log Analysis, Artifact Management, Failure Triage, Flake Detection, Regression Testing, Release Gates, Staged Rollouts, Jira, Agile

Software Engineer

Sept 2021 – Dec 2022

Infor

Hyderabad, India

- Developed and maintained automated test frameworks and developer tooling using **Python, C++, Java, and shell scripting** for enterprise and device-connected application workflows across integration, system, and end-to-end validation layers.
- Built reusable automation utilities for API validation, environment provisioning, log collection, test execution, failure reproduction, and debugging of software components across Linux-based development and test environments.
- Worked with Linux internals including processes, networking, storage, system services, and systemd-based components to troubleshoot application startup, resource, connectivity, and integration failures.
- Developed repeatable CI/CD validation workflows using Git, Jenkins, Maven, and containerized environments, publishing execution artifacts and enforcing automated regression checks before release promotion.
- Implemented failure-triage processes and regression analysis to identify flaky tests, isolate environmental failures, distinguish product defects from test-infrastructure problems, and improve overall signal reliability.
- Created automated monitoring and test-reporting mechanisms for pass rates, regression stability, execution trends, and release-readiness metrics, providing visibility into software quality across release cycles.
- Collaborated with Software Engineers, QA, Product, and infrastructure teams to improve application testability, investigate complex system failures, and support staged releases and production-readiness validation.
- Prepared technical documentation, test procedures, troubleshooting guides, failure-analysis reports, release notes, and repeatable validation instructions used by engineering and QA teams.

Environment: Python, C++, Java, Bash, Linux, systemd, Networking, Storage, REST APIs, Git, Jenkins, Maven, Docker, CI/CD, Integration Testing, System Testing, End-to-End Testing, Regression Infrastructure, Failure Triage, Flake Analysis, Release Validation

Associate Engineer QA

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed automated validation frameworks for web, mobile, and device-integrated applications using Java, Python, Selenium WebDriver, Appium, Cucumber BDD, and reusable test infrastructure.
- Supported hardware-software integration testing across Android and iOS devices, including application provisioning, configuration, device connectivity, automated execution, log collection, and repeatable regression workflows.
- Created reusable automation scripts for environment setup, API validation, device-state verification, log capture, test-data management, and failure reproduction across development and QA environments.
- Performed system-level debugging across mobile applications, backend services, APIs, device connectivity, networking, and application logs to isolate defects spanning software and hardware-dependent workflows.
- Integrated automated suites with Jenkins, Maven, Git, and CI/CD pipelines to support repeatable test execution, build verification, regression validation, and release-readiness checks.
- Tracked regression pass rates, recurring failures, unstable scenarios, and environmental defects; collaborated with engineering teams to improve flaky-test reliability and strengthen quality signals.
- Worked with Software Engineers, QA leads, Product teams, and external testing groups to execute standardized test procedures, reproduce failures, validate fixes, and support production release activities.
- Prepared detailed test procedures, defect reports, debugging evidence, execution summaries, release-validation results, and troubleshooting documentation supporting repeatable engineering workflows.

Environment: Java, Python, Selenium WebDriver, Appium, Android, iOS, Device Testing, Hardware-Software Integration, REST APIs, Linux, Git, Jenkins, Maven, CI/CD, Log Capture, Failure Triage, Regression Testing, System Testing, End-to-End Testing, Release Readiness

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.